EXP NO:4 To Count the frequency of occurrence of a word in a body of text is often needed during text processing.

AIM:

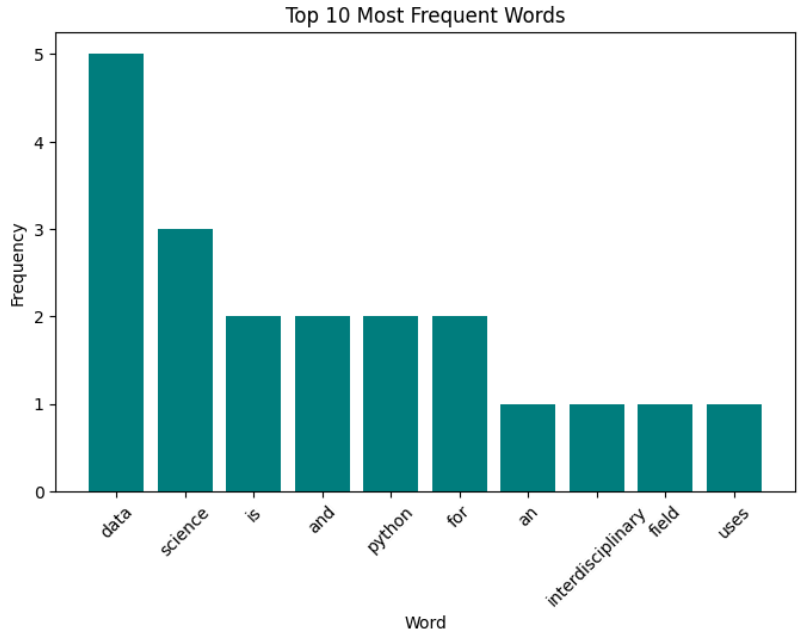
To count the frequency of occurrence of a word in a body of text, which is often needed during text processing.

PROGRAM:

python

```
import re
import pandas as pd
import matplotlib.pyplot as plt
from collections import Counter

# Body of text
text = """Data science is an interdisciplinary field. Data science uses
statistics, programming and domain knowledge to extract insights from data.
Python is the most popular language for data science because Python has
powerful libraries for data analysis and visualization."""

# Convert to lowercase and remove punctuation
clean_text = re.sub(r'[^\w\s]', '', text.lower())

# Split the text into words
words = clean_text.split()
print("Total words:", len(words))

# Count the frequency of every word
word_freq = Counter(words)
print(word_freq)

# Frequency of a particular word
search_word = "data"
print("Frequency of '{}':".format(search_word), word_freq[search_word])

# Top 5 most common words
print("Top 5 words:", word_freq.most_common(5))

# Convert to a DataFrame
df = pd.DataFrame(word_freq.most_common(10), columns=["Word", "Frequency"])
df

# Bar chart of the top 10 words
plt.figure(figsize=(8, 5))
plt.bar(df["Word"], df["Frequency"], color="teal")
plt.title("Top 10 Most Frequent Words")
plt.xlabel("Word")
plt.ylabel("Frequency")
plt.xticks(rotation=45)
plt.show()
```

OUTPUT:



RESULT:

Thus the program to count the frequency of occurrence of a word in a body of text was executed successfully.